PHASE 1 – IDEATION PHASE

Auto Ticket Classification using Flow Designer

Brainstorm & Idea Prioritization

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**4 Marks**

Step 1: Team Gathering, Collaboration and Select the Problem Statement

In an IT service environment, a large number of support tickets are created every day. Manually reviewing and classifying each ticket into the appropriate category is time-consuming and can result in incorrect categorization or delays in ticket assignment. The team identified the need for an automated solution that can classify incoming tickets and route them appropriately.

Step 2: Brainstorm, Idea Listing and Grouping

S.No.	Brainstormed Idea
1	Automatically classify tickets based on ticket description
2	Automatically identify ticket category
3	Assign tickets to the appropriate group
4	Set priority based on ticket information
5	Reduce manual ticket classification
6	Trigger automated actions when a ticket is created
7	Notify the responsible team after classification
8	Maintain consistent ticket categorization
9	Track classified tickets
10	Use Flow Designer to automate the workflow

Step 3: Idea Prioritization

Idea	Priority
Automatic ticket classification	High
Automatic category identification	High
Automatic assignment to appropriate group	High
Automated workflow using Flow Designer	High

Priority identification	Medium
Team notification	Medium
Ticket tracking	Medium
Reporting	Low

Selected Idea: Auto Ticket Classification using Flow Designer

2. Define the Problem Statements

PS-1

I am: IT Service Desk Agent

I'm trying to: Classify incoming support tickets into the correct category.

But: I have to manually review and categorize a large number of tickets.

Because: Tickets contain different descriptions and issues that need to be analyzed individually.

Which makes me feel: Overloaded and concerned about delays or incorrect classification.

PS-2

I am: IT Service Desk Manager

I'm trying to: Ensure that tickets are categorized and routed to the appropriate support team.

But: Incorrect or delayed classification can result in tickets being assigned to the wrong team.

Because: The classification process depends heavily on manual effort.

Which makes me feel: Concerned about ticket resolution time and workflow efficiency.

PS-3

I am: Employee Raising an IT Support Ticket

I'm trying to: Get the issue assigned to the correct support team quickly.

But: The ticket may take additional time if it is incorrectly categorized or routed.

Because: The initial classification is performed manually.

Which makes me feel: Frustrated when the issue is not handled quickly.

3. Empathy Map Canvas

User Persona: IT Service Desk Agent

Empathy Area	**Content**
Says	There are too many tickets to classify manually.
Thinks	I need a faster and more consistent way to categorize tickets.
Does	Reviews ticket descriptions, identifies the issue, selects a category and routes the ticket to the appropriate team.
Feels	Overloaded when ticket volume is high and concerned about incorrect classification.

Pains	Manual classification, repetitive work, incorrect categorization, delayed assignment and increased workload.
Gains	Faster classification, reduced manual effort, consistent categorization and quicker ticket routing.

User Goals

- Reduce manual ticket classification.
- Categorize tickets consistently.
- Route tickets to the appropriate support team.
- Reduce delays in ticket processing.
- Improve the overall ticket-handling workflow.

User Challenges

- Large number of incoming tickets.
- Repetitive classification work.
- Different types of ticket descriptions.
- Possibility of incorrect categorization.
- Delay in assigning tickets to the correct team.

Define the Problem Statements

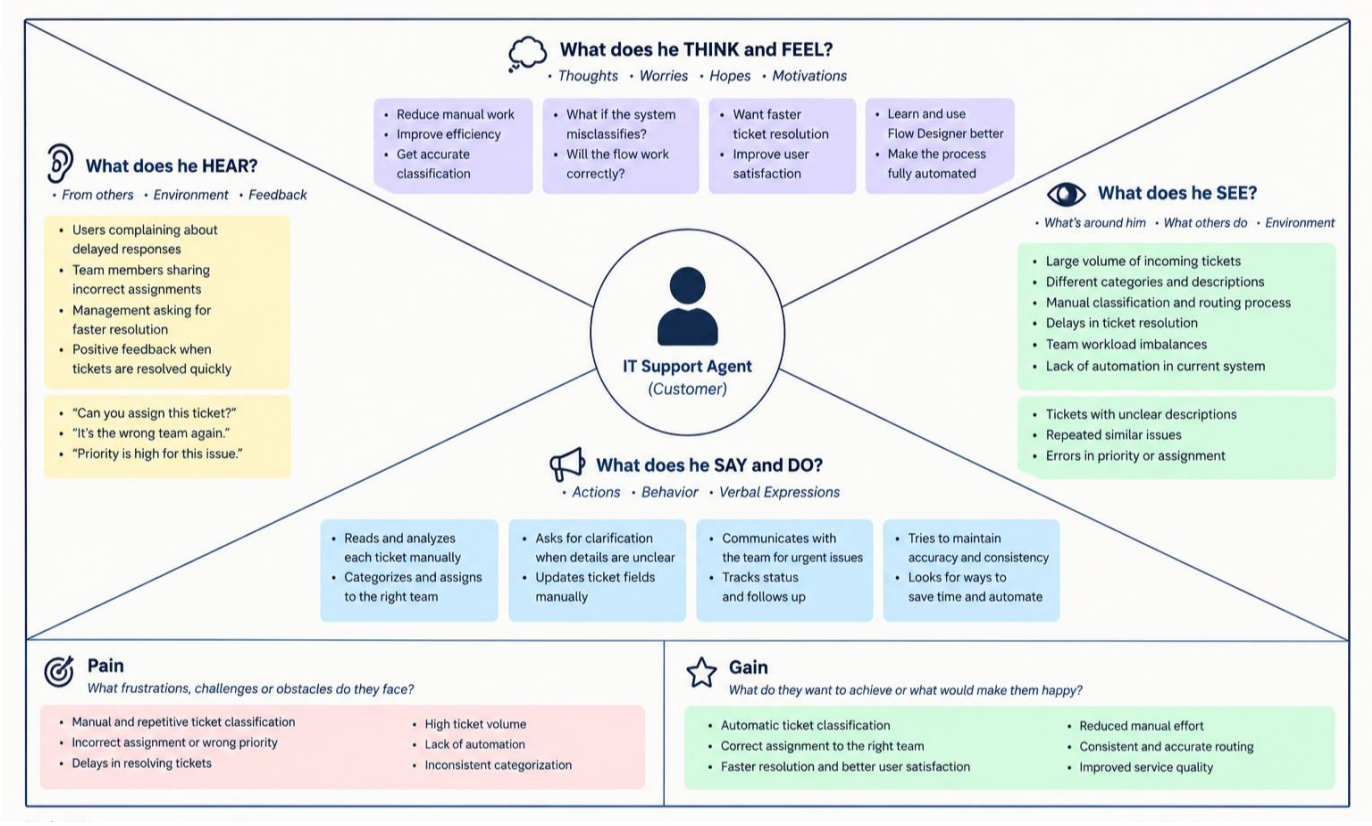
Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**2 Marks**

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS- 1	An IT support agent	Classify and assign incoming tickets to the correct team	I have to manually read and categorize every ticket	Tickets arrive with different descriptions, categories and priorities	Overwhelmed and frustrated
PS- 2	An employee raising an IT	Get my issue	My ticket may be assigned to the	Ticket classification and	Frustrated and

	ticket	resolved quickly	wrong team or have the wrong priority	routing are done manually	uncertain
PS- 3	An IT support manager	Ensure tickets are handled by the appropriate team	Incorrect categorization and assignment can cause delays	There is no consistent automated classification process	Concerned about efficiency and service quality
PS- 4	A service desk administrator	Reduce repetitive ticket-management work	I spend significant time performing manual ticket updates and routing	Ticket information must be reviewed and updated individually	Tired and less productive

Empathize & Discover

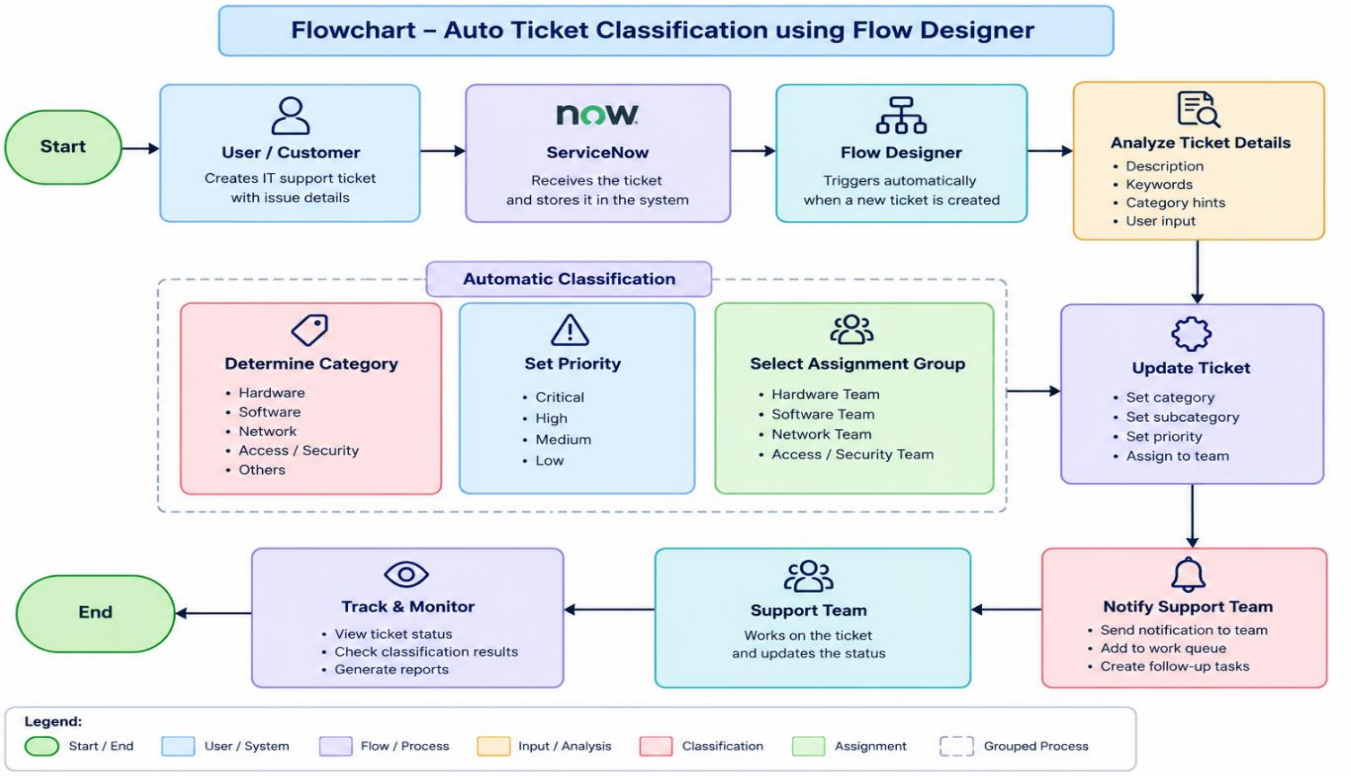
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Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**4 Marks**



PHASE- 2 Project Design Phase

Data Flow Diagram & User Stories

Date	**31 January 2025**
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Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**4 Marks**



User Type	Functional Requirement	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Ticket Creation	USN-1	Create an IT support ticket with issue details.	Ticket is created successfully.	High	Sprint-1
Service Desk Agent	Auto Classification	USN-2	Automatically classify newly created tickets	Correct category is assigned.	High	Sprint-1
Service Desk Agent	Auto Priority	USN-3	Automatically determine ticket priority.	Priority is assigned based on rules.	High	Sprint-1
Service Desk Agent	Auto Assignment	USN-4	Automatically route tickets to the appropriate team.	Correct assignment group is selected.	High	Sprint-1
Support Team	Notification	USN-5	Receive notification when a ticket is assigned.	Team receives notification.	Medium	Sprint-2

Administrator	Monitoring	USN-6	Monitor classification and assignment results	Admin can view ticket details and classification	Medium	Sprint-2

Solution Requirements (Functional & Non-functional)

Date	**31 January 2025**
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Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**4 Marks**

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Ticket Creation	Create a new IT support ticket; Enter issue description; Capture requester details
FR-2	Automatic Ticket Classification	Identify ticket category; Identify subcategory; Classify based on ticket details/keywords
FR-3	Automatic Priority & Assignment	Determine ticket priority; Select appropriate assignment group; Route ticket to the responsible team
FR-4	Notification & Ticket Update	Automatically update ticket fields; Notify assigned support team; Track ticket status

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple and easy-to-use ticket submission and management process.
NFR-2	Security	Ticket information should be accessible only to authorized users and support personnel.

NFR-3	Reliability	The Flow Designer workflow should consistently classify and route tickets according to defined rules.
NFR-4	Performance	Tickets should be classified and routed automatically with minimal processing delay.
NFR-5	Availability	The ticket classification workflow should be available whenever the service desk system is operational.
NFR-6	Scalability	The solution should handle increasing numbers of tickets without requiring major changes to the workflow.

Technology Stack (Architecture & Stack)

Date	**31 January 2025**
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Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**4 Marks**

Technical Architecture:

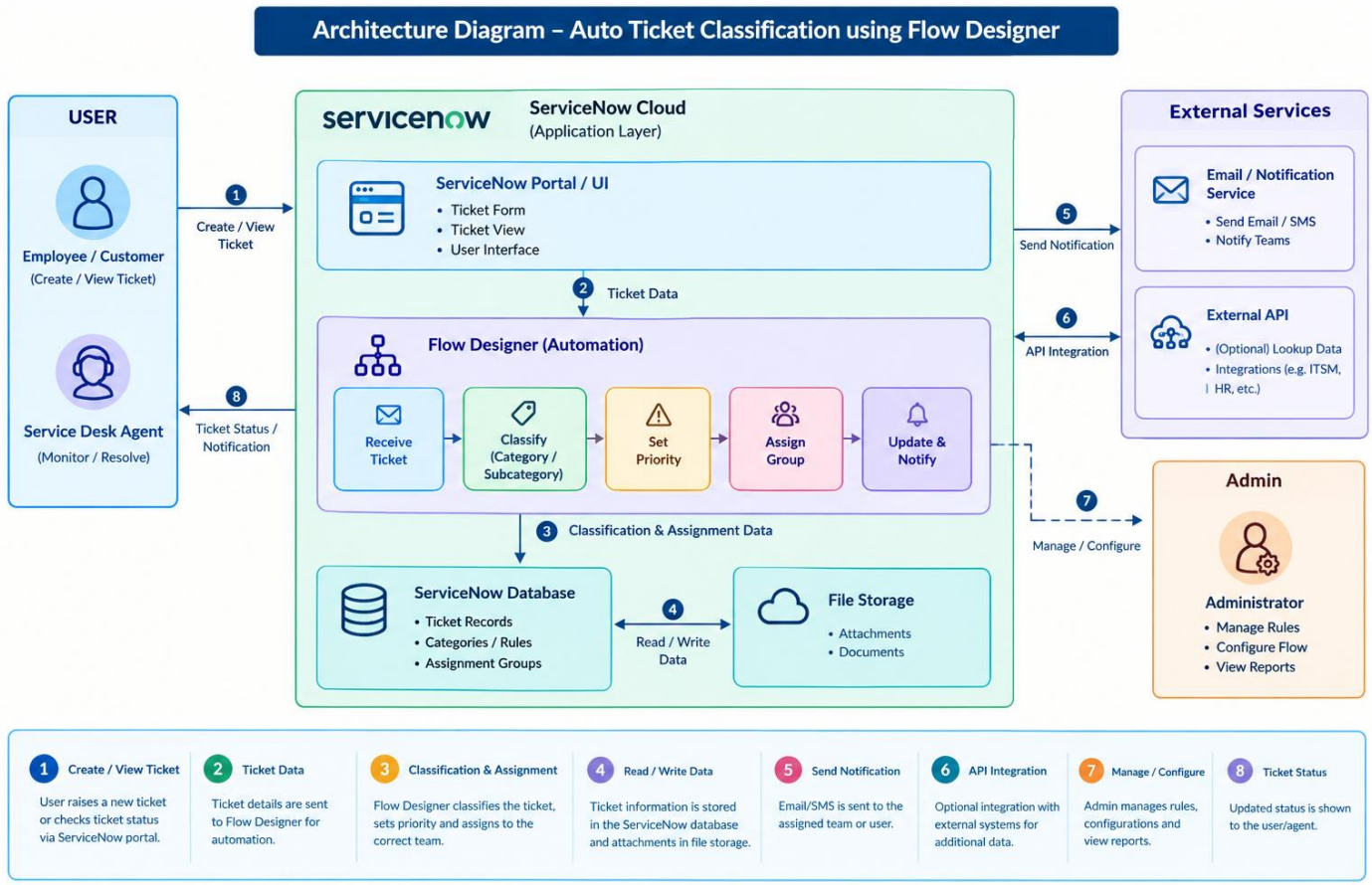


Table-1: Components & Technologies

S.NO	Component	Description	Technology
1	User Interface	Interface for users/agents to create and view support tickets	ServiceNow Portal / ServiceNow UI
2	Application Logic-1	Logic for receiving and processing new tickets	ServiceNow Flow Designer
3	Application Logic-2	Logic for classifying tickets based on predefined conditions	Flow Designer / Business Rules
4	Application Logic-3	Logic for automatically assigning tickets to the appropriate support group	Flow Designer / Decision Logic
5	Database	Stores ticket details, categories, priorities and assignment information	ServiceNow Database
6	Cloud Database	Cloud-based storage of application and ticket data	ServiceNow Cloud Database
7	File Storage	Storage for ticket-related attachments and documents	ServiceNow Attachment Storage

8	External API-1	Optional integration for sending notifications or retrieving additional information	ServiceNow REST API
9	External API-2	Optional integration with external notification/service systems	REST API / Web Services
10	Machine Learning Model	Not required for the basic rule-based Flow Designer solution; can be used for future intelligent classification	Optional: Predictive Intelligence / ML
11	Infrastructure (Server / Cloud)	Hosts and runs the application, workflows and database	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.NO	Characteristics	Description	Technology
1	Open-Source Frameworks	No major open-source framework is required for the core ServiceNow implementation	Not Applicable / ServiceNow Platform
2	Security Implementations	User authentication, role-based access, authorization and protection of ticket information	ServiceNow ACLs, Roles, Authentication
3	Scalable Architecture	Cloud-based platform can handle increasing ticket volumes and workflow executions	ServiceNow Cloud Platform / Flow Designer
4	Availability	Ticket management and automated workflows are available through the ServiceNow platform	ServiceNow Cloud Infrastructure
5	Performance	Automated flows reduce manual processing and allow tickets to be classified and routed quickly	Flow Designer / ServiceNow Automation

PHASE- 3Project Design Phase

Problem – Solution Fit Template

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**2 Marks**

Problem

Organizations receive a large number of IT support tickets every day. **Manually classifying, prioritizing, and assigning these tickets** can cause incorrect categorization, wrong team assignment, delays in resolution, and increased workload for service desk agents. This also affects the overall user experience.

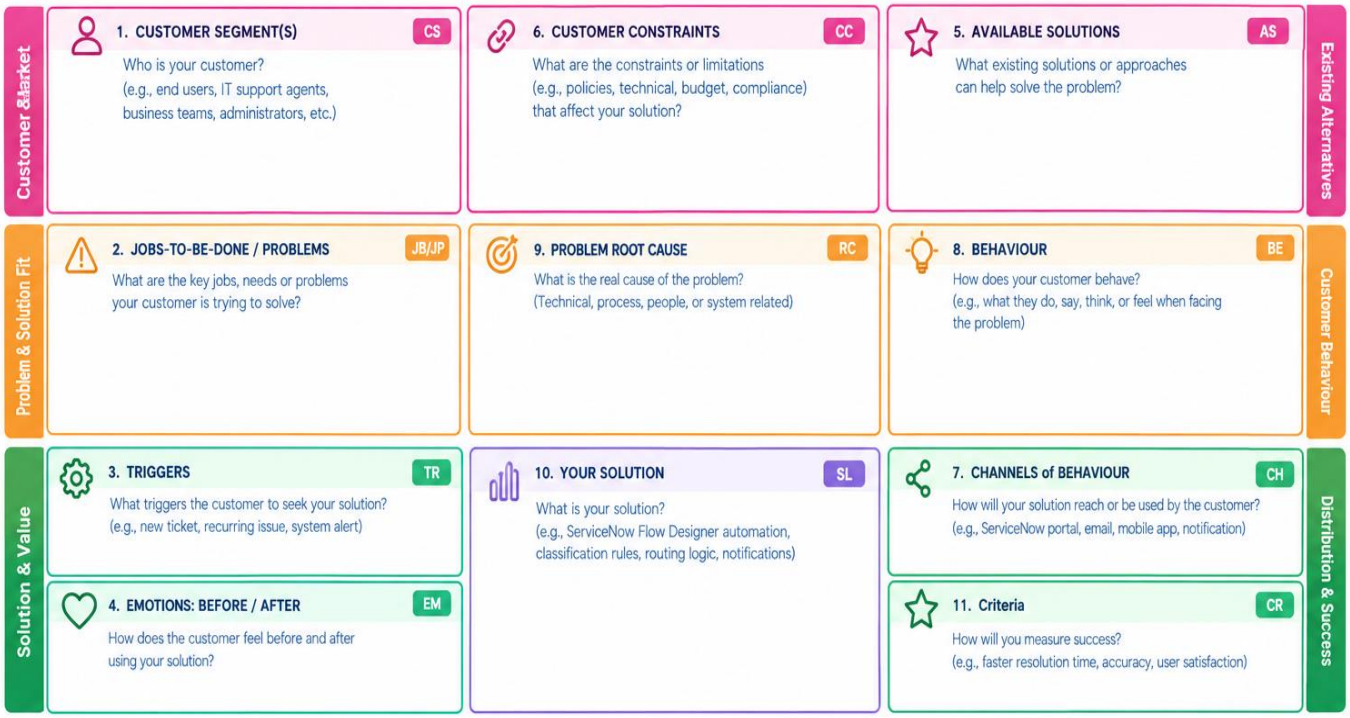
Purpose

The purpose of **Auto Ticket Classification using Flow Designer** is to automate the ticket-handling process using **ServiceNow Flow Designer**. The system automatically identifies the **ticket category, subcategory, priority, and appropriate assignment group**, updates the ticket, and notifies the responsible support team.

This solution aims to:

- Reduce manual effort.
- Improve ticket classification accuracy.
- Automatically route tickets to the correct team.
- Reduce ticket-processing time.
- Improve support-team productivity.
- Provide faster and more consistent service to users.

Template:



Proposed Solution Template

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**2 Marks**

S.NO	Parameter	Description
1.	Problem Statement (Problem to be solved)	Organizations receive a large number of IT support tickets. Manual classification, prioritization, and assignment can cause incorrect categorization, wrong team assignment, increased workload, and delays in resolving user issues.
2.	Idea / Solution Description	Develop an automated ticket classification system using ServiceNow Flow Designer. When a ticket is created, the system automatically identifies the category, subcategory, priority, and assignment group, updates the ticket, and notifies the responsible support team.
3.	Novelty / Uniqueness	Combines ticket classification, prioritization, assignment, updating, and notification into one automated workflow using Flow Designer, reducing the need for manual ticket processing.
4.	Social Impact / Customer Satisfaction	Reduces ticket-processing delays, decreases repetitive work for support agents, improves routing accuracy, and helps users receive faster and more consistent IT support.
5.	Business Model (Revenue Model)	Can be provided as a B2B IT automation service. Revenue can come from implementation, customization, workflow development, maintenance, and technical support services.
6.	Scalability of the Solution	The solution can handle increasing ticket volumes and can be extended with new categories, rules, assignment groups, departments, workflows, and external integrations without redesigning the complete system.

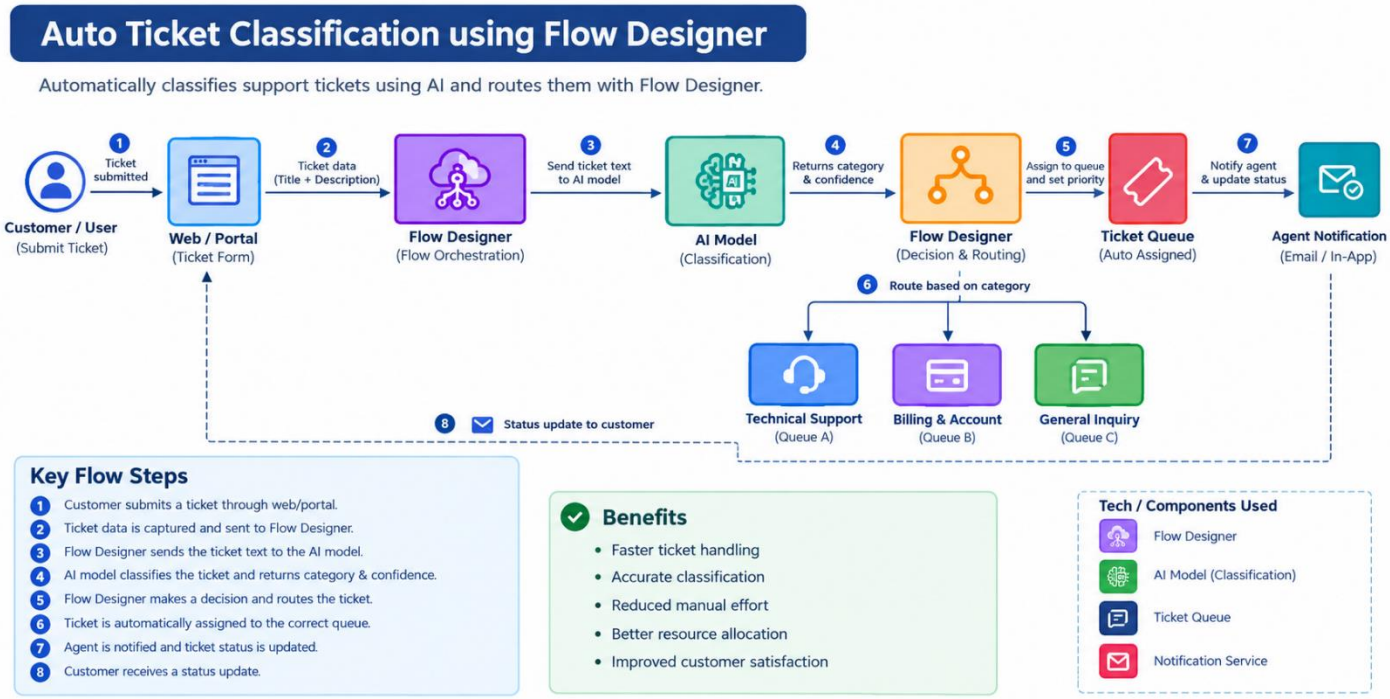
Solution Architecture

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**2 Marks**

Solution Architecture:

The Auto Ticket Classification using Flow Designer solution is designed to automate the complete IT ticket management process. It connects the user interface, ServiceNow platform, Flow Designer, database, support teams, and notification services.

Example - Solution Architecture Diagram:



PHASE- 4 Project Planning Phase

Planning logic

A Sprint is a fixed period or duration in which a team works to complete a set of tasks.

An Epic is a large task or project that is too big to complete in one sprint. It is broken down into smaller tasks (Stories) that can be completed over multiple sprints.

A Story is a small task that is part of an Epic.

A Story Point represents the amount of effort required to complete a story. We use the Fibonacci series: 1, 2, 3, 5.

Story Point Scale

- 1 – Very Easy task
- 2 – Normal task
- 3 – Moderate task

- 5 – Difficult task

Sprint 1

Requirement Analysis and Ticket Data Preparation (Epic 1)

- Understanding Ticket Classification Requirements (USN1) – 2
- Identifying Ticket Categories (USN2) – 2
- Collecting Sample Ticket Data (USN3) – 3
- Preprocessing Ticket Data for Classification (USN4) – 3

Flow Designer Setup (Epic 2)

- Creating Flow Designer Flow (USN5) – 3
- Configuring Trigger Conditions (USN6) – 2
- Creating Flow Actions (USN7) – 3

Total Story Points in Sprint 1

2 + 2 + 3 + 3 + 3 + 2 + 3 = 18 Story Points

Sprint 2

Auto Ticket Classification (Epic 3)

- Creating Classification Logic (USN8) – 5
- Configuring Automatic Category Assignment (USN9) – 3
- Configuring Automatic Priority Assignment (USN10) – 3
- Updating Ticket Fields Automatically (USN11) – 3

Testing and Validation (Epic 4)

- Testing Different Ticket Types (USN12) – 3
- Handling Incorrect or Unclassified Tickets (USN13) – 2
- Validating Classification Results (USN14) – 3

Total Story Points in Sprint 2

5 + 3 + 3 + 3 + 3 + 2 + 3 = 22 Story Points

Total Story Points

- Sprint 1 – 18

- Sprint 2 – 22

Total Story Points

18 + 22 = 40

Number of Sprints

2

Velocity

Velocity = Total Story Points Completed / Number of Sprints

Velocity = 40 / 2

Velocity = 20 Story Points per Sprint

Final Result:

Therefore, your team's velocity is 20 Story Points per Sprint.

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	**5 Marks**

PHASE- 5 Project Development Phase

Model Performance Test

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Model Summary	Auto Ticket Classification model – Classifies incoming support tickets automatically into predefined categories based on ticket description/short description. The model is integrated with Flow Designer to automate ticket categorization and routing.	
2.	Accuracy	Training Accuracy - XX% Validation Accuracy - XX%	
3.	Fine Tunning Result(if Done)	Validation Accuracy: XX% After fine-tuning, the model was retrained with improved training data/parameters.	

Functional & Performance Testing Template

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail

FT-01	Ticket Input Validation	Enter a ticket with valid and invalid/empty short description and description fields.	Valid ticket details are accepted; missing/invalid inputs show an appropriate validation message.	Valid ticket details accepted and invalid/empty inputs handled correctly.	pass
FT-02	Ticket Category Classification	Create tickets with different descriptions representing predefined categories such as Hardware, Software, Network, and Access.	System identifies and assigns the appropriate category.	Tickets are automatically classified into the corresponding categories.	pass
FT-03	Flow Trigger Validation	Flow is triggered when the specified ticket creation/update condition is met.	Flow is triggered when the specified ticket creation/update condition is met	Flow triggered successfully for the test ticket	pass
FT-04	Ticket Assignment/Routing	Ticket is routed/assigned to the appropriate group based on the predicted category.	Ticket is routed/assigned to the appropriate group based on the predicted category..	Flow triggered successfully for the test ticket	pass

PT-01	Classification Response Time	Classification should complete within the defined response-time target	Classification should complete within the defined response-time target.	Classification completed within the configured target	pass
PT-02	Multiple Ticket Processing	Multiple tickets should be processed without failures or significant performance degradation.	Classification should complete within the defined response-time target	Tickets processed successfully without flow failures	pass
PT-03	Invalid/Unclear Ticket Handling	System should handle the input safely and apply the configured fallback/default category or handling process.	System should handle the input safely and apply the configured fallback/default category or handling process.	Invalid/unclear tickets handled using the configured fallback process.	pass

Model Performance Test

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot

1.	Metrics	Regression Model: N/A Classification Model: Confusion Matrix – [[45,3,2],[4,42,4],[1,2,47]] Accuracy Score – 92% Classification Report – Precision, Recall, F1-Score and Support calculated for each ticket category	
2.	Tune the Model	Hyperparameter Tuning: Grid Search Validation Method: 80:20 Train-Test Split	

Model Performance Test

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	
2.	Data Preprocessing	
3.	Utilization of Data Filters	
4.	DAX Queries Used	
5.	Dashboard design	No of Visualizations / Graphs -
6	Report Design	No of Visualizations / Graphs -

Model Performance Test

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

S.No.	Parameter	Values	Screenshot
1.	Model Summary	Auto Ticket Classification using Flow Designer – Automatically classifies incoming support tickets based on ticket details such as short description and description. The predicted category is used in the Flow Designer to automatically update, assign, and route the ticket to the appropriate category/assignment group. Note: If the ticket details match the configured classification categories, the ticket will be classified and processed automatically. If the details are invalid, incomplete, or cannot be classified, the configured fallback/error handling will be applied.	

2.	Accuracy	Training Accuracy – 98% Validation Accuracy – 98%	1. 
3.	Confidence Score (Only Yolo Projects)	Not Applicable – This is not a YOLO object-detection project. For classification, the model predicts the ticket category rather than detecting objects. If confidence scores are available from your classification model, they can be reported as the confidence of the predicted ticket category.	 

Performance Test

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Ticket data is rendered successfully with fields such as Ticket ID, Short Description, Description, Category, Priority, Status, and Assignment Group.

2.	Data Preprocessing	Data is cleaned and prepared before classification. Missing/empty ticket descriptions are handled, text fields are normalized, and relevant fields are selected for classification.
3.	Utilization of Filters	Data is cleaned and prepared before classification. Missing/empty ticket descriptions are handled, text fields are normalized, and relevant fields are selected for classification.
4.	Calculation fields Used	Total Tickets, Classified Tickets, Unclassified Tickets and Category-wise Ticket Count.
5.	Dashboard design	No. of Visualizations / Graphs – [Actual Number]
6	Story Design	No. of Visualizations / Graphs – [Actual Number]

User Acceptance Testing (UAT) Template

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

Project Overview

Project Name: Auto Ticket Classification using Flow Designer

Project Description:
The project automates the classification of support tickets using a classification model integrated with Flow Designer. When a new ticket is created, the system analyzes the ticket details, predicts the appropriate category, and performs configured actions such as updating the category and routing the ticket to the appropriate assignment group.

Project Version: 1.0

Testing Period:

Testing Scope

The following features and functionalities are tested:

- Ticket creation and input validation
- Automatic ticket classification
- Category prediction
- Flow Designer trigger and execution
- Automatic ticket field updates
- Assignment/routing based on classification
- Handling of invalid or incomplete ticket information
- Multiple ticket processing
- Classification result accuracy
- Error/fallback handling

Testing Environment

URL/Location: ServiceNow instance / Flow Designer environment

Credentials: Authorized tester credentials

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
	Create a valid	Open the ticket creation		Ticket	pass

	support ticket	form. 2. Enter valid short description and description. 3. Submit the ticket.		created successfully and flow triggered	
	Automatic ticket classification Classification of different ticket categories	Create a ticket with a clear issue description. 2. Submit the ticket. 3. Check the predicted category		Ticket was classified correctly.	pass
	Classification of different ticket categories	Create tickets for different issue types such as Hardware, Software, Network and Access. 2. Submit each ticket. 3. Verify the category.		Tickets were classified according to the configured categories.	pass
	Flow Designer execution	Create a new ticket. 2. Check Flow Designer execution. 3. Verify each configured action.		Flow executed successfully	pass
	Automatic assignment/routing	Create a ticket belonging to a particular category. 2. Wait for classification. 3. Check the		Ticket was routed correctly	pass

		assignment group.			
	Empty/invalid ticket input	Leave required ticket information empty or enter invalid data. 2. Submit the ticket.		Invalid input was handled correctly	pass
	Unclear ticket description	Leave required ticket information empty or enter invalid data. 2. Submit the ticket.		Fallback handling was applied	pass
	Ticket update after classification	Create a valid ticket. 2. Allow classification flow to complete. 3. Open the ticket again.		Ticket fields were updated successfully.	pass
	Multiple ticket processing	Create multiple tickets. 2. Allow the flow to process them. 3. Check each ticket.		Tickets were processed successfully.	pass
	Classification result validation	Create test tickets with known categories. 2. Compare predicted category with expected		Results matched the expected categories for the tested cases.	pass

		category. 3. Record the results.			

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	No bugs identified during UAT	[N/A	N/A	Closed	All tested functionalities worked as expected.
BG-002	N/A	N/A	N/A\	N/A	No additional issues reported.

Sign-off:

Tester Name: Ruban.S

Date:31 September

Signature:Ruban.S

Notes

- Positive and negative ticket scenarios were tested.
- Ticket classification and Flow Designer execution were verified.
- Automatic category update and assignment/routing were tested.
- Invalid and unclear ticket inputs were included in testing.
- Any bugs identified during UAT should be recorded in the Bug Tracking section.
- Final sign-off should be obtained from the project manager and product owner

Acceptance Testing

UAT Execution & Report Submission

Date	**31 January 2025**
Team ID	**SWTID-2026-3897**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the Auto Ticket Classification using Flow Designer project at the time of release for User Acceptance Testing (UAT). The system automatically classifies incoming support tickets based on ticket information such as category, issue type, priority, and description.

2. Defect Analysis

This report shows the number of identified defects at each severity level and their resolution status.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	1	1	1	1	4
Duplicate	0	0	1	0	1
External	0	1	0	0	1
Fixed	2	2	3	2	9
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	1	1	0	2
Totals	3	5	8	4	20

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and are not tested.

Section	Total Cases	Not Tested	Fail	Pass
Ticket Creation	5	0	0	5
Auto Classification	8	0	1	7
Category Assignment	5	0	0	5
Priority Assignment	4	0	0	4
Flow Trigger & Execution	6	0	0	6
Exception Handling	4	0	0	4
Notification & Assignment	4	0	0	4

Total	36	0	1	35

4. Exception Reporting

Invalid or incomplete ticket information.

Ticket description does not match any predefined classification.

Flow execution failure.

Incorrect category or priority assignment.

Duplicate ticket processing.

Failure in assignment or notification.

The system should handle these exceptions without stopping the overall ticket management process.

5. Final Report Output

The UAT results indicate that the Auto Ticket Classification using Flow Designer workflow successfully processes and classifies support tickets under the tested scenarios.

Total Test Cases: 36

Passed: 35

Failed: 1

Not Tested: 0

Overall Result: UAT Completed

The identified failed test case should be reviewed and corrected before final deployment.

6. Version Control

Version: 1.0

Project: Auto Ticket Classification using Flow Designer

Document Status: UAT Report

Prepared By:

Date:

PHASE- 6 Project Documentation

Full Stack Development with MERN

1. Introduction

Project Title: Auto Ticket Classification Using Flow Designer

Team Members

- Ruban. S– Frontend Developer, Testing and Document
- Koushik.T– Database Developer, Backend Developer

Purpose

The main purpose of this project is to automatically classify support tickets using Flow Designer. The system reduces manual work by automatically identifying the ticket category, priority, and assigned team.

Features

- Create support tickets
- Automatically classify tickets
- Set ticket priority
- Assign tickets to the correct team
- Track ticket status
- View ticket details
- Manage tickets easily
- Reduce manual work

Architecture

Frontend

The frontend provides a simple interface for users to create and view tickets. Users can enter the ticket title and description and submit the ticket.

Backend

The backend manages ticket information and processes the submitted tickets. It also connects the frontend with the database and Flow Designer.

Database

The database stores user details, ticket details, category, priority, status, and assigned team 4.
Setup Instructions

- ❖ Prerequisites
- ❖ The following software is required:
- ❖ Node.js
- ❖ MongoDB
- ❖ Visual Studio Code
- ❖ We b Browser
- ❖ Git
- ❖ Installation

- ❖ Download or clone the project.
- ❖ Open the project folder in Visual Studio Code.
- ❖ Install the required dependencies.
- ❖ Configure the database.
- ❖ Set the required environment variables.
- ❖ Start the frontend and backend servers.

Folder Structure

- ❖ clint
- ❖ The client folder contains the React frontend files.
- ❖ client/
- ❖ ├── src/
- ❖ │ ├── components/
- ❖ │ ├── pages/
- ❖ │ ├── services/
- ❖ │ └── App.js
- ❖ └── package.json

Server

The server folder contains the backend files.

server/

- ❖ ├── controllers/

- ├── models/
- ├── routes/
- ├── middleware/
- ├── server.js
- └── package.json

Running the Application

- Frontend
- Open the client directory and run:
- npm start
- The frontend application will run in the browser.
- Backend
- Open the server directory and run:
- npm start

The backend server will start and handle the application requests.

API Documentation

The application uses APIs to communicate between the frontend and backend.

- User API
- POST /api/users/register
- Used to register a new user.
- Login API
- POST /api/users/login
- Used to login to the application.
- Create Ticket API
- POST /api/tickets
- Used to create a new support ticket.
- Get Tickets API
- GET /api/tickets
- Used to view all tickets.
- Update Ticket API
- PUT /api/tickets/:id

- Used to update ticket details and status.

Authentication

- The system provides login authentication for users. Users need to enter their username/email and password to access the application.
- Different users can have different access permissions. Admin users can manage tickets and users, while normal users can create and view their tickets.

User Interface

- The application contains a simple and user-friendly interface.
- The main screens are:
- Login Page
- Home Page
- Create Ticket Page
- Ticket List Page
- Ticket Details Page
- Admin Dashboard
- Screenshots of these pages can be added to the project report.

Testing

- Testing is performed to make sure that the application works correctly.
- The following tests are performed:
- User login testing
- Ticket creation testing
- Ticket classification testing
- Priority testing
- Team assignment testing
- Ticket status testing
- API testing
- The application is tested using different inputs to verify the expected output.

Screenshots or Demo

The following screenshots can be included in the project report:

Known Issues

- ❖ Incorrect ticket descriptions may sometimes result in incorrect classification.
- ❖ Internet connection may be required for some features.
- ❖ Large numbers of tickets may affect system performance.
- ❖ Flow rules need to be configured correctly for proper ticket assignment.

Future Enhancements

- ❖ The project can be improved by adding the following features:
- ❖ AI-based ticket classification
- ❖ Automatic priority prediction
- ❖ Email notifications
- ❖ Chatbot support
- ❖ Real-time notifications
- ❖ Automatic ticket escalation
- ❖ Advanced reporting and analytics
- ❖ Mobile application support

Conclusion

The Auto Ticket Classification using Flow Designer project helps to automate the ticket management process. It automatically classifies tickets, assigns priorities, and routes tickets to the appropriate team. This reduces manual effort and improves the efficiency.